

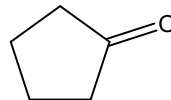
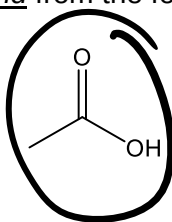
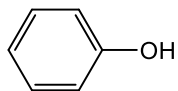
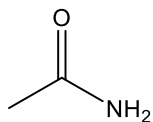
Quiz 1

Name: _____

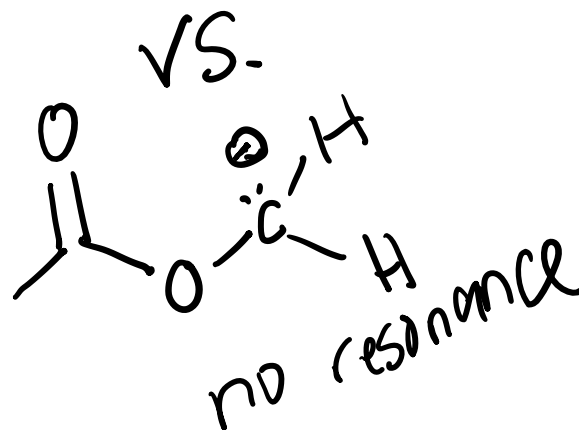
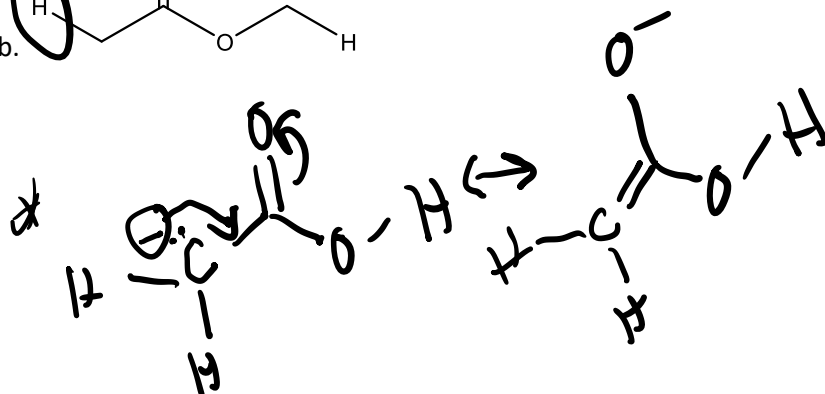
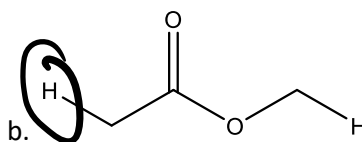
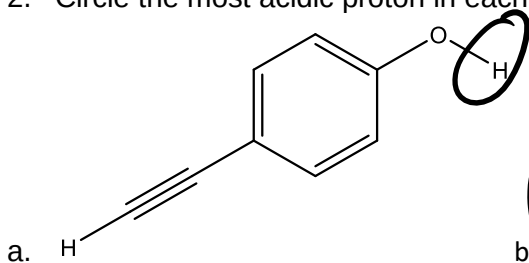
Last

First

1. Circle the most acidic compound from the following set.



2. Circle the most acidic proton in each of the following species.



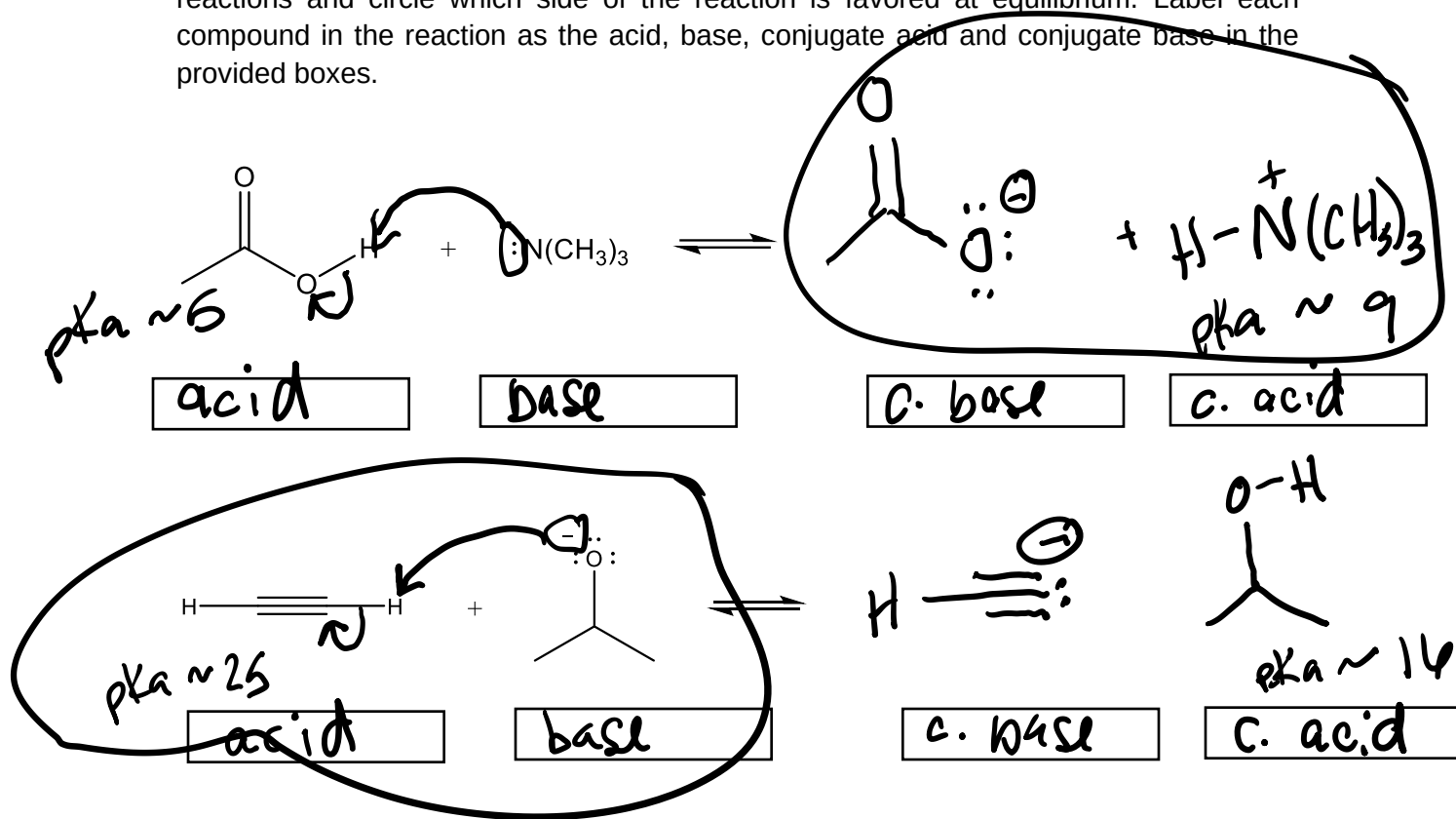
Quiz 1

Name: _____

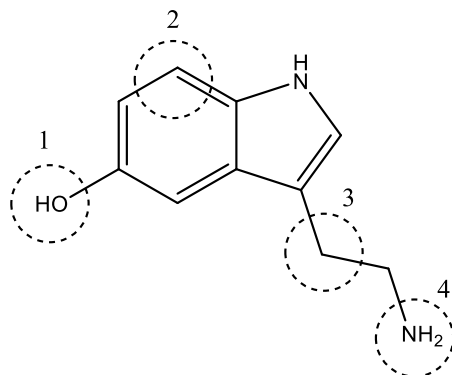
Last

First

3. Draw curved arrows showing the formation of products for the following acid/base reactions and circle which side of the reaction is favored at equilibrium. Label each compound in the reaction as the acid, base, conjugate acid and conjugate base in the provided boxes.



4. Below is the structure of serotonin. Identify the hybridization for each of the indicated atoms.



1. sp^3
2. sp^2
3. sp^3
4. sp^3

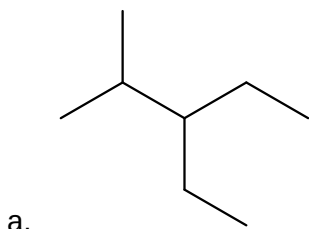
Quiz 1

Name: _____

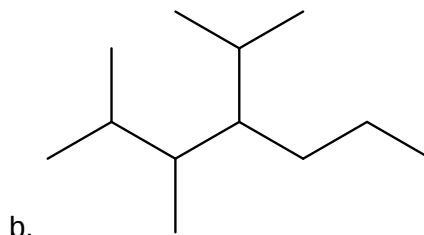
Last

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5. Give the IUPAC name for each of the following compounds on the line below:



3-ethyl-2-methylpentane

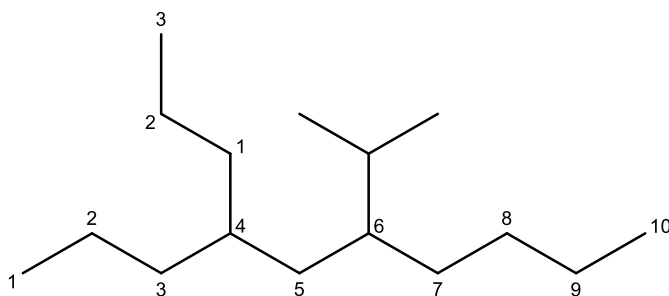


b.

4-isopropyl-2,3-dimethylheptane OR

4-(1-methylethyl)-2,3-dimethylheptane

6. Draw a line structure for 6-isopropyl-4-propyldecane.

7. Provide a **brief** explanation for the following:

a. Hydroiodic acid is more acidic than hydrochloric acid

I is lower on the periodic table than Cl. Therefore, the conjugate base I⁻ of HI is more stabilized due to size.

b. Comparing the two anions, NH₂⁻ is a stronger base than OH⁻

N is less electronegative than O. Therefore NH₂⁻ is less stable than OH⁻ with a negative charge.